Linux basics commonds

1. NAVIGATION COMMANDS (Move around system)

🔹 **pwd → Print current directory**

```
pwd
```

👉 Shows where you are

🔹 **ls → List files**

```
ls
ls -l
ls -a
```

👉 View files, hidden files, permissions

🔹 **cd → Change directory**

```
cd Desktop
cd ..
cd /home/kali
```

👉 Move between folders

🟢 2. FILE & DIRECTORY MANAGEMENT

🔹 **mkdir → Create folder**

```
mkdir test
```

🔹 **rmdir → Remove empty folder**

```
rmdir test
```

🔹 **rm → Delete files/folders**

```
rm file.txt
rm -r folder
rm -rf folder
```

⚠️ -rf = dangerous (force delete)

🔹 **cp → Copy files**

```
cp file.txt backup.txt
cp -r folder1 folder2
```

🔹 **mv → Move/rename**

```
mv file.txt newname.txt
mv file.txt /home/kali/
```

🟢 3. FILE VIEWING & EDITING

- **cat → Show file content**

```
cat file.txt
```

- **less → Scroll file**

```
less file.txt
```

👉 Better for large files

- **head → First lines**

```
head file.txt
```

- **tail → Last lines**

```
tail file.txt
tail -f log.txt
```

👉 Used for logs (real pentesting use)

- **nano → Edit file**

```
nano file.txt
```

🟢 4. SEARCHING (VERY IMPORTANT)

- **find → Find files**

```
find / -name file.txt
```

- **grep → Search text**

```
grep "password" file.txt
```

- **Combine:**

```
cat file.txt | grep admin
```

👉 Used to find credentials/logs

🟢 5. PERMISSIONS & OWNERSHIP

- **chmod → Change permissions**

```
chmod +x script.sh
chmod 777 file.txt
```

🔹 **chown → Change owner**

```
chown user file.txt
```

👉 Permissions meaning:

- r = read
- w = write
- x = execute

🟢 6. USER & SYSTEM COMMANDS

🔹 **whoami**

```
whoami
```

👉 Shows current user

🔹 **id**

```
id
```

👉 Shows user ID (used in privilege escalation)

🔹 **sudo**

```
sudo apt update
```

👉 Run as admin

🔹 **passwd**

```
passwd
```

👉 Change password

🟢 7. PROCESS MANAGEMENT

🔹 **ps**

```
ps aux
```

👉 Show running processes

🔹 **top**

```
top
```

👉 Live system processes

🔹 **kill**

kill 1234

👉 Stop process

⚫ 8. NETWORK COMMANDS (IMPORTANT FOR HACKING)

🔹 **ifconfig / ip a**

ip a

👉 Show IP address

🔹 **ping**

ping google.com

👉 Check connectivity

🔹 **netstat**

netstat -tuln

👉 Show open ports

🔹 **ss**

ss -tuln

👉 Modern alternative to netstat

🟢 9. PACKAGE MANAGEMENT (INSTALL TOOLS)

🔹 **apt (Kali/Ubuntu)**

sudo apt update
sudo apt install nmap

🟢 10. ARCHIVE & FILE TRANSFER

🔹 **zip / unzip**

zip file.zip file.txt
unzip file.zip

🔹 **tar**

tar -cvf file.tar folder/
tar -xvf file.tar

🟢 11. PERMISSION ANALYSIS (VERY IMPORTANT FOR PENTESTING)

🔹 **Check permissions:**

ls -l

Example:

-rwxr-xr-x

👉 Means:

- Owner → full access
- Others → limited

🟢 12. HISTORY & SHORTCUTS

🔹 **history**

history

🔹 **clear**

clear